DO NOW, AO1 REVISION

Specific heat capacity (RP)

Energy
Paper 1, Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the three variables that a change in thermal energy depends on.**	Mass, specific heat capacity and change in temperature	☐
2	**What does Δθ represent?**	Change in temperature	☐
3 ★	**State the standard unit of specific heat capacity.**	Joule per kilogram per degree Celsius (J/kg °C)	☐
4 ★	**Define specific heat capacity.**	The energy required to raise the temperature of 1 kg of a substance by 1 °C.	☐
5	**A material has a high specific heat capacity. Is it hard or easy to heat up, and why?**	Hard. It takes a lot of energy to raise the temperature of 1 kg by 1 °C.	☐
6 ★	**State the three measurements needed to find the specific heat capacity of a substance.**	Energy supplied, mass and temperature rise	☐
7	**Name the piece of equipment used to measure temperature.**	A thermometer	☐
8	**Name the piece of equipment used to measure mass.**	A balance	☐
9	**Explain why a Bunsen burner is not used when measuring specific heat capacity.**	The energy it supplies cannot be measured easily.	☐
10 ★	**Explain why an electrical heater is used when measuring specific heat capacity.**	The energy supplied can be calculated from potential difference, current and time.	☐
11	**State the equation linking change in thermal energy, mass, specific heat capacity and temperature change.**	change in thermal energy = mass × specific heat capacity × temperature change ($\Delta E = m\ c\ \Delta\theta$)	☐
12 ★	**When measuring specific heat capacity, explain why the block is wrapped in insulation.**	To reduce energy transfer to the surroundings.	☐

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	QUESTION	ANSWER	✓
13 ★	**Explain why a measured specific heat capacity is usually higher than the true value.**	Some of the energy supplied is transferred to the surroundings, not the block.	☐

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